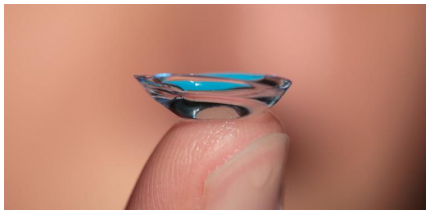
Hydrogels

BT 5011 Biomaterials Engineering

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Hydrogels



Hydrogel contact lens



Gel ice packs

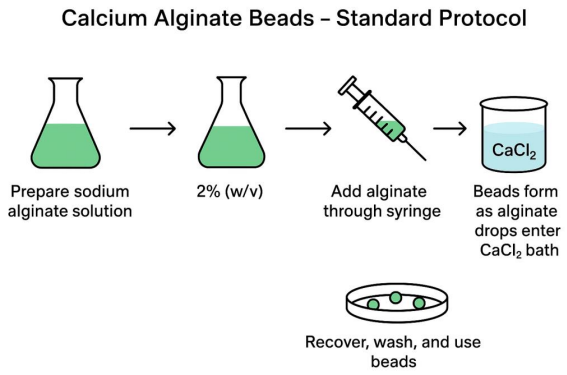


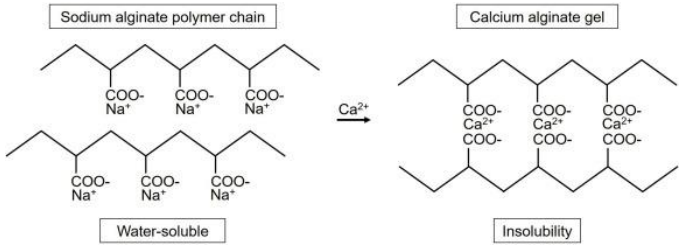
Hydrogel wound dressings

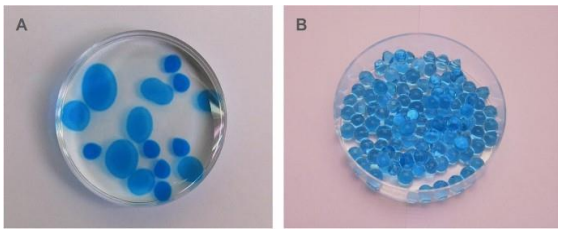
What are hydrogels

- Definition: Interconnected polymer network built using hydrophilic or amphiphilic monomers that are rendered insoluble due to the presence of crosslinks
- Key properties of a hydrogel
  - Three-dimensional crosslinked polymer network
  - Hydrophilic groups (–OH, –COOH, $–CONH_2$, etc.)
  - Absorb and retain large amounts of water (typically ≥10% of its weight)
  - Swelling Without Dissolution
  - Generally used above their glass transition temperature (Tg)

What are hydrogels







What are hydrogels

- Different physical forms of hydrogel

(a) solid moulded forms (e.g., soft contact lenses)

(b) pressed powder matrices (e.g., pills or capsules for oral ingestion)

(c) microparticles (e.g., as bio adhesive carriers or wound treatments)

(d) coatings (e.g., on implants or catheters; on pills or capsules; or coatings on the inside capillary wall in capillary electrophoresis)

(e) membranes or sheets (e.g., as a reservoir in a transdermal drug delivery patch; or for 2D electrophoresis gels)

(f) encapsulated solids (e.g., in osmotic pumps)

(g) liquids (e.g., that form gels on heating or cooling)

Why hydrogels

- Structural and compositional similarity to extracellular matrix
- Extensive framework for cell proliferation and survival
- Possible engineering for
  - Cell attachment – incorporation of cell attachment moieties
  - Molecular response – stimuli responsiveness
  - Structural integrity – mechanical properties
  - Biodegradability
  - Biocompatibility
  - Solute transport

Solute transport

- Critical design parameter for hydrogels
- Mass transfer parameter
  - Nutrients
  - Gases
  - Waste products
  - Bioactive agents such as growth factors
- Convection doesn’t play a big role
  - Why?
  - Are there exceptions?
    - Large micropores
    - Force flow conditions
- Diffusion is the driving transport phenomenon

Solute transport

- If diffusion is primary for transport, what factors could contribute to small molecules and protein diffusion?
  - Mesh size
  - pH
  - Temperature
- pH dependent hydrogels could be prepared to exhibit zero-order or near zero order drug release
  - What does this mean?
  - Why is this important?

Hydrogel swelling

- One or more highly electronegative atoms which results in charge asymmetry favoring hydrogen bonding with water
- Because of their hydrophilic nature dry materials absorb water
- By definition, water must constitute at least 10% of the total weight (or volume) for a materials to be a hydrogel
- When the content of water exceeds 95% of the total weight (or volume), the hydrogel is said to be superabsorbent

Hydrogel swelling

- Highly swollen hydrogel
  - Cellulose derivatives
  - Poly(vinyl alcohol)
  - Poly(ethylene glycol)
  - What do all these have in common?
    - Lot of OH or =O groups to interact with acidic environments → hydrophilic → swelling
- Moderately swollen hydrogels
  - PHEMA and derivatives
- Copolymerization can be used for getting desired swelling properties

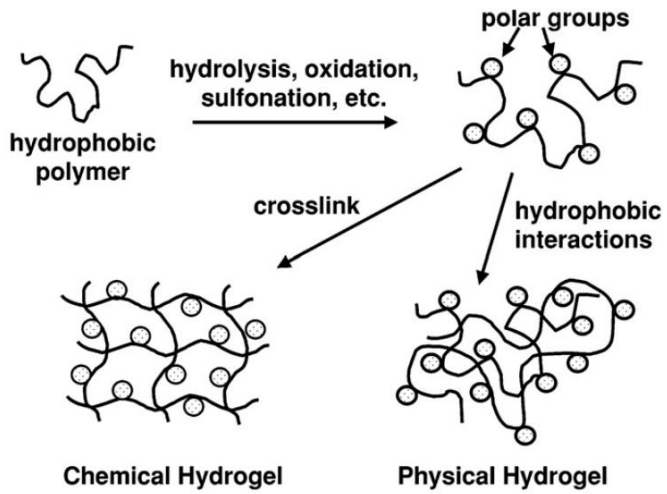
Polymers used for hydrogel preparation

Natural polymers and their derivatives (±*crosslinkers*)
Anionic polymers: HA, alginic acid, pectin, carrageenan, chondroitin sulfate, dextran sulfate
Cationic polymers: chitosan, polylysine
Amphipathic polymers: collagen (and gelatin), carboxymethyl chitin, fibrin
Neutral polymers: dextran, agarose, pullulan

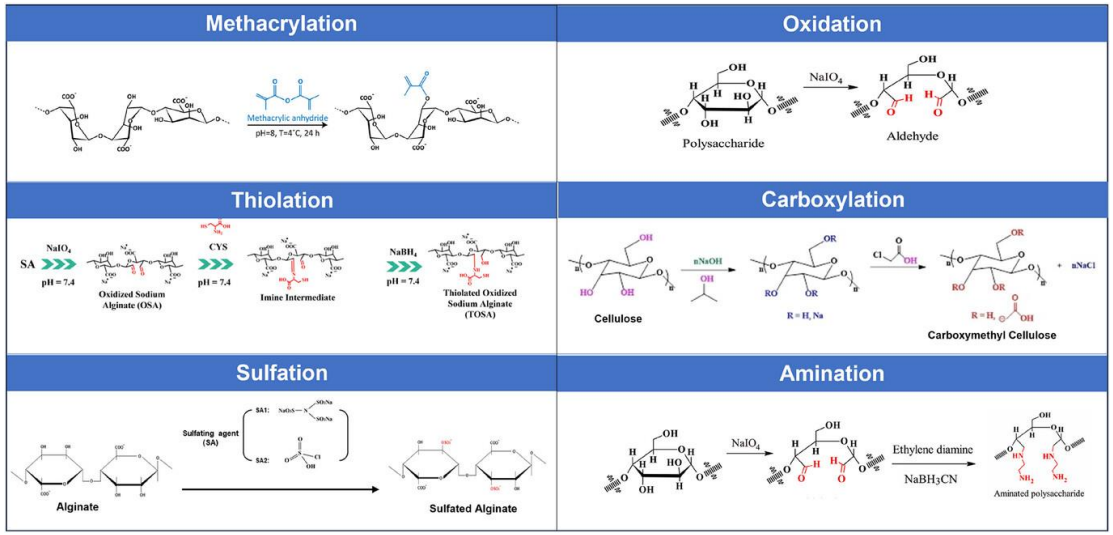
Synthetic polymers (±*crosslinkers*)
Polyesters: PEG-PLA-PEG, PEG-PLGA-PEG, PEG-PCL-PEG, PLA-PEG-PLA, PHB, P(PF-co-EG)±acrylate end groups, P(PEG/PBO terephthalate)
Other polymers: PEG-bis-(PLA-acrylate), PEG±CDs, PEG-g-P(AAm-co-Vamine), PAAm, P(NIPAAm-co-AAc), P(NIPAAm-co-EMA), PVAc/PVA, PNVP, P(MMA-co-HEMA), P(AN-co-allyl sulfonate), P(biscarboxy-phenoxy-phosphazene), P(GEMA-sulfate)

Combinations of natural and synthetic polymers
P(PEG-co-peptides), alginate-g-(PEO-PPO-PEO), P(PLGA-co-serine), collagen-acrylate, alginate-acrylate, P(HPMA-g-peptide), P(HEMA/Matrigel®), HA-g-NIPAAm

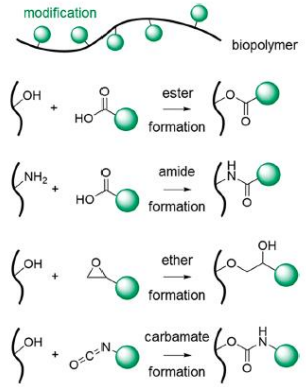
Synthesis of hydrogels

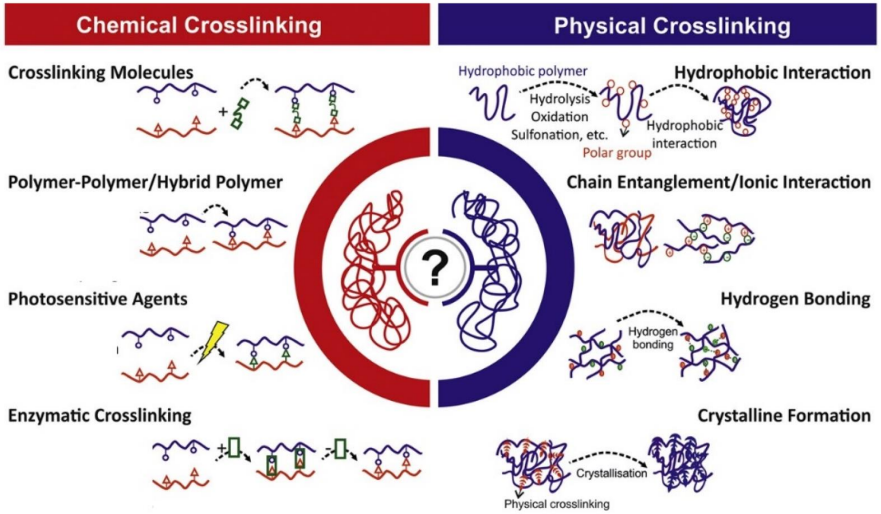


Polymer modifications



Synthesis of hydrogels





Engineering hydrogels

- Design considerations
  - Biocompatibility
  - Vascularization
  - Degradation
  - Macroenvironment
  - Microenvironment
- Bioactive hydrogels
  - Immobilized biological motifs
    - RGD peptides
  - Soluble growth factors
    - Heparan sulfate proteoglycans (HSPGs)

Next class

- Activity: Polymer combination and the crosslinking strategy
- Types of hydrogels
- Examples and applications

Hydrogels

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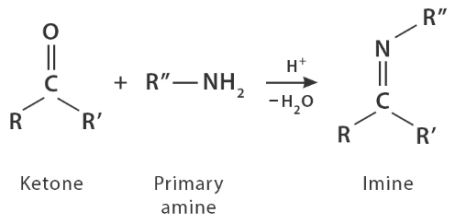
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Last class

- What are hydrogel?
- Why hydrogels?
- Importance of their water holding ability, swelling, solute transport
- Polymers used for hydrogel preparation
- Synthesis methods
- Polymer modifications
- Physical and chemical crosslinking
- Activity

Chemical crosslinking strategies

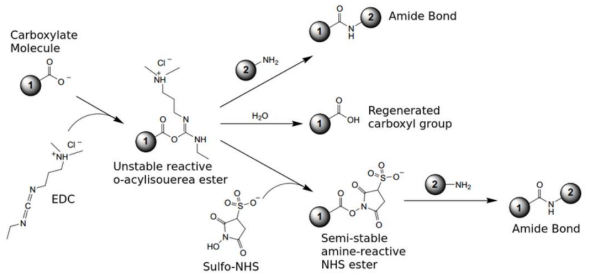
- Imine bond (Schiff's base)
- Between an <u>aldehyde group</u> and a <u>primary amine group</u>



- Characteristics: Reversible under physiological conditions (Self-healing), unstable under acidic environment

Chemical crosslinking strategies

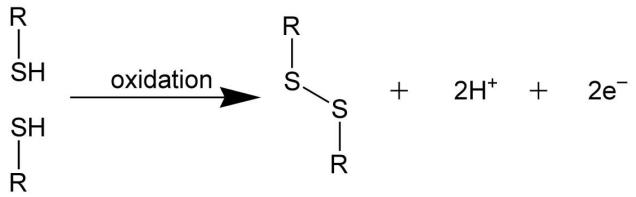
- Amide bond (Carbodiimide chemistry)
- Between a carboxylic acid group and a primary amine group



- Characteristics: Stable linkage, biocompatible

Chemical crosslinking strategies

- Disulfide bond
- Between <u>thiol groups</u>

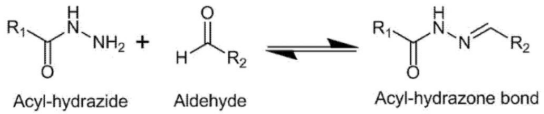


- Characteristics: Redox-responsive crosslinking, biocompatible, reversible

Chemical crosslinking strategies

- Acyl hydrazone bond

- Between <u>acyl hydrazide group</u> and an <u>aldehyde/ketone group</u>

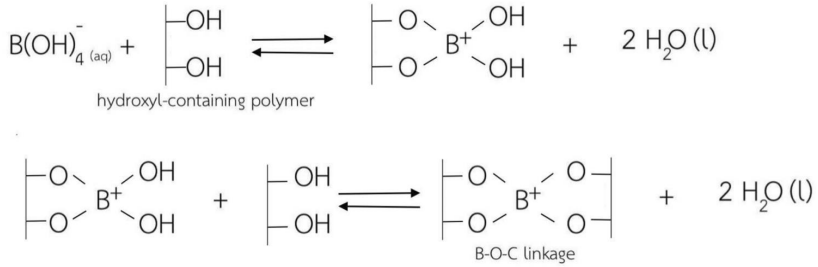


- Characteristics: Reversible, acid pH labile

Chemical crosslinking strategies

- Boronate ester bond

- Between <u>hydroxyl groups</u>

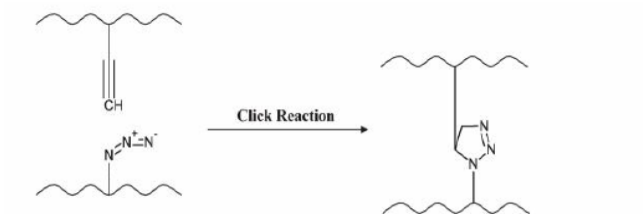


- Characteristics: Reversible, self-healing, cross-linker required

Chemical crosslinking strategies

- Click chemistry

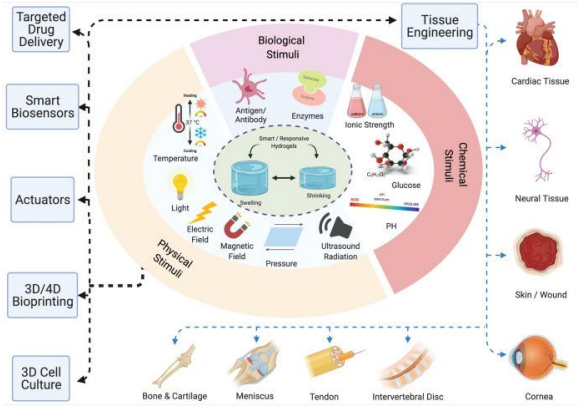
- Between azide and terminal alkyne moieties



- Characteristics: cross-linker required, stable, irreversible, fast, in situ gelation

Stimuli-responsive hydrogels

- Sensitive to external stimuli such as pH, temperature, light, electrical field, etc. achieved by using stimuli-responsive crosslinkers
  - Shrink, swell, bend or degrade in response to the stimuli

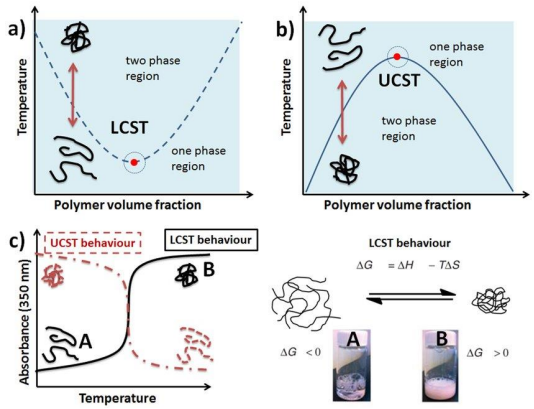


Stimuli-responsive hydrogels

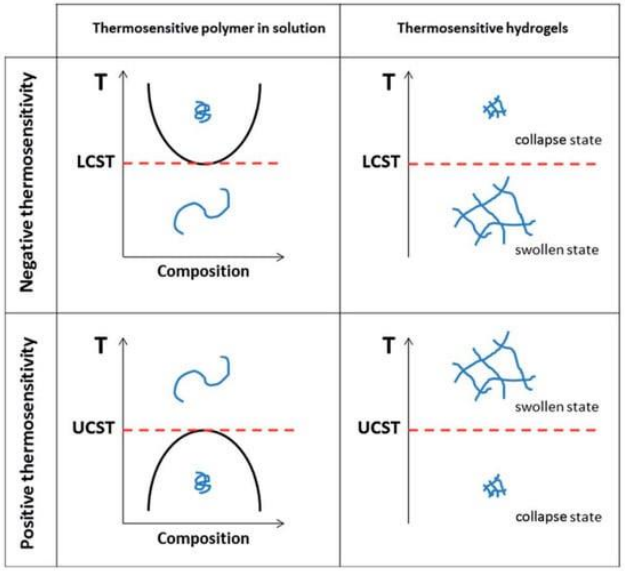
Environmental stimuli	Hydrogel	Mechanism	Applications
Electrical signal	Polyelectrolyte hydrogel	Change in charge distribution causes swelling and drug release	Actuator, artificial muscle, on/off drug release.
Thermal	Thermo-responsive hydrogel, e.g., poly (Nisopropylacrylamide)	Change in polymer–polymer and polymer–water interactions cause swelling and drug release	On/off drug release, squeezing device
pH	Acidic or basic hydrogel e.g. PAA, PDEAEM	Ionization of polymer chain upon pH change; pH change causes swelling and release of drug	pH-dependent oral drug delivery
Magnetic fields	Magnetic particles dispersed in the hydrogel matrix	Applied magnetic field causes pores in gel and swelling followed by drug release	Controlled drug delivery while the magnetic particles used form medical therapy.
Ionic strength	Ionic hydrogel	Change in concentration of ions inside the gel cause swelling and release of drug	Biosensor for glucose, used for medical therapy
Chemical	Hydrogel containing electron-accepting groups e.g. Chitosan -PEO	Formation of charge-transfer complex cause swelling and release of drug	Controlled drug delivery
Enzyme–substrate	Hydrogel containing immobilized enzymes	Product of enzymatic conversion causes swelling and release of drug	Modulated drug release in the presence of a specific antigen; sensor for immunoassay and antigen.
Light	Copolymer of PNIPAAm	Temperature change via the incorporated photosensitive molecules; dissociation into ion pairs by UV irradiation	Optical switches, ophthalmic drug delivery

Thermoresponsive hydrogels

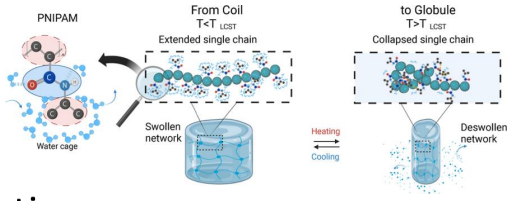
- Extensively studied class of smart hydrogels
- Depends on the phase transition in response to temperature.



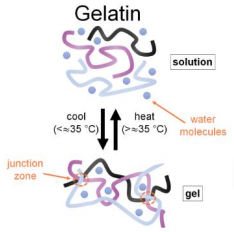
Thermoresponsive hydrogels



- PNIPAAm

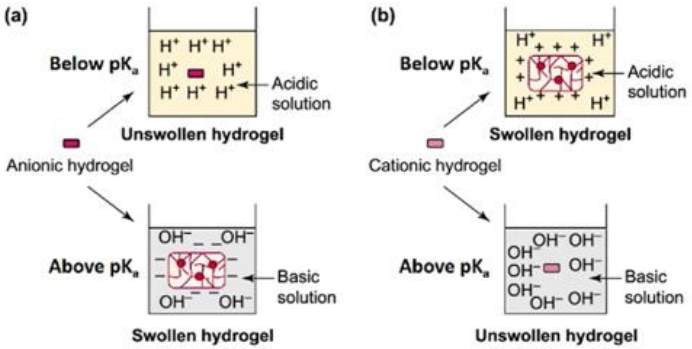


- Gelatin



pH-responsive hydrogels

- Made from monomers with ionic groups or by crosslinking polyelectrolytes with acidic or basic side groups.
- Eg: Poly(acrylamide), poly(acrylic acid), poly(methacrylic acid)



Photo/light responsive hydrogels

- Stimulated by light to induce changes in the physical or chemical property of the hydrogel
- Prepared by functionalization of polymer backbone or incorporation of photo responsive groups
- The given optical signal is captured by the photosensitizer that converts the light into chemical signals which changes the property of the hydrogel

Hydrogels: Advantages

- Non-thrombogenic
  - Non-ionic hydrogels used for blood contacting applications
  - Heparinized hydrogels show promise
- Biocompatible
- Good transport of nutrients to cells and products from cells
- May be easily modified with cell adhesion ligands
- Can be injected in vivo as a liquid that gels at body temperature

Hydrogels: Disadvantages

- Difficult to handle
- Mechanically weak
- Difficult to sterilize

Hydrogel in Tissue Engineering

- Scaffolds
  - Mimics ECM
  - Cells can be adhered to the matrix or suspended within the gel
    - Which is preferred and why?
  - Biodegradable vs. non-biodegradable
- Barriers
  - Avoids post-operative re-stenosis and thrombosis
  - Prevent platelets, coagulation factors and plasma proteins from contacting vascular wall

Hydrogel in Tissue Engineering

- Drug delivery
  - Controlled and smart drug release
  - Engineering delivery kinetics by altering degree of swelling, crosslinking density, and degradation rate
- Cell encapsulation
  - Provides immuno-isolation while diffusion of other products happens
  - Artificial pancreas
  - Microencapsulation using alginate-Ca^{2+}